CS 549: Machine Learning Final Project Report
Baseball Player Stat Projector

Matthew Sprague

***Abstract*—This project introduces an innovative strategy for predicting Major League Baseball (MLB) player performance statistics using machine learning techniques. I utilized Random Forest and XGBoost models to predict essential batting statistics for upcoming seasons. The algorithm uses historical player data from 2013 to 2023 to forecast metrics including batting average (AVG), on-base percentage (OBP), slugging percentage (SLG), home runs (HR), runs batted in (RBI), weighted on-base average (wOBA), and weighted runs created plus (wRC+).**

Keywords—baseball analytics, player projection, machine learning, sports statistics

I. Introduction

Major League Baseball has seen significant changes in recent decades, due to the widespread use of advanced analytics in numerous aspects of the sport. Data-driven approaches have become crucial to baseball operations, ranging from player evaluation and scouting to in-game decisions. One crucial area that continues to provide a difficulty for teams and analysts is player performance projection—the ability to effectively predict how a player will perform in future seasons.

Traditional projection systems, such as Marcel, PECOTA, Steamer, and ZiPS, have utilized weighted averages, regression to the mean, and aging curves to forecast player performance. While these methods have shown reasonable accuracy, they often fail to capture the complicated, non-linear developments in player performance.

This project hopes to solves these problems by creating a machine learning-based projection system that uses Random Forest and XGBoost based models, to predict essential batting statistics. The algorithm analyzes historical data from the 2013-2023 MLB seasons to find trends in player development and performance changes over time.

II. Related Work

Marcel, created by Tom Tango, provided a simple baseline based on a 3-year weighted average, regression to the mean, and fundamental aging effects. Despite its simplicity, Marcel has proven to be unexpectedly effective, and it now serves as a standard against which other systems are judged.

PECOTA (Player Empirical Comparison and Optimization Test Algorithm), developed by Nate Silver, generates estimates based on comparable player identification and advanced aging curves. The method generates a variety of alternative performance scenarios using past player comparisons and percentile-based results.

III. Methodology

To provide multi-year player performance estimates, the MLB player projection system uses a multifaceted strategy that combines data collecting,

feature engineering, model training, and cascade predictions.

A. Data Collection

I used the pybaseball API, which provides complete baseball statistics by scraping existing database websites like FanGraphs, to collect player performance data for MLB seasons ranging from 2013 to 2023. Players who made fewer than 100 plate appearances (PA) in a season were eliminated to provide a large enough sample size for relevant analysis.

The dataset includes 15 features for each player-season:

- Basic metrics: Age, Games (G), Plate Appearances (PA)
- Traditional statistics: Batting Average (AVG), On-Base Percentage (OBP), Slugging Percentage (SLG)
- Counting statistics: Home Runs (HR), Runs Batted In (RBI)
- Advanced metrics: Walk Rate (BB%), Strikeout Rate (K%), Isolated Power (ISO), Batting Average on Balls In Play (BABIP), Weighted On-Base Average (wOBA), Weighted Runs Created Plus (wRC+), Wins Above Replacement (WAR)

For model training, I built feature-target pairs in which the features were a player's statistics from season t and the targets were the same player's performance in season t+1. This method enabled the models to identify patterns in year-by-year performance changes.

B. Models Used

I used two learning techniques in my projection system:

Random Forest Regression: Random Forest Regression is a decision tree ensemble that reduces overfitting by aggregating bootstraps and randomizing features. Random forests excel at capturing nonlinear relationships and feature interactions.

XGBoost: XGBoost is a gradient boosting framework that creates trees in sequence to fix faults in prior models. XGBoost uses regularization terms to prevent overfitting and has regularly outperformed in machine learning competitions.

I trained individual Random Forest and XGBoost models for each target statistic (AVG, OBP, SLG, HR, RBI, wOBA, wRC+), totaling 14 models (2 methods x 7 statistics).

The Random Forest implementation used 100 estimators with scikit-learn's default hyperparameters, whereas the XGBoost models used 100 estimators with the squared error objective function. While hyperparameter tweaking could possibly improve outcomes, I discovered that these settings provided the right combination of performance and computing efficiency.

My approach incorporates cascading forecasts for multi-year estimates. Rather than training distinct models for two-, three-, and longer time frames, we use year t+1 forecasts to anticipate year t+2.

The process works as follows:

1. Use actual statistics from year t to forecast performance in year t+1.
2. For year t+1, average the predictions made by both models (Random Forest and XGBoost).
3. Use these averaged projections as characteristics to forecast the year t+2.
4. Continue this technique for the succeeding years.

This cascade method enables the system to predict performance over numerous seasons without the need for individual models for each projection timeframe. It also naturally reflects the increase in uncertainty in longer-term predictions.

I used Python's pickle package to create model persistence, avoiding the need to retrain models for each user. Trained models are serialized and saved on disk so that they can be loaded fast to generate

predictions. This method dramatically improves the system's response time and user experience. It allows for efficient reproducibility and replicability of the results of the system.

C. Implementation

This system was designed to be user focused, allowing the user to select any baseball player that comes to mind and immediately view their predicted statistics for a specified number of future years. The user is first prompted to name a player. Then the starting year that they want to start projecting stats from (this should be the current year). The user is then shown a table of predicted stats for each season of the specified number of future seasons. This table details the stats predicted by both models: Random Forrest and XGBoost. Following this table, the user is then asked if they would like to visualize these projections. The user can select any target stat they would like to visualize, and if they proceed, a graph of the specified projected target statistics will be displayed. This was implemented using matplotlib.

IV. Experimental Results and Analysis

A. Training Performance

During training, models were evaluated using Mean Squared Error (MSE), which was calculated based on training data predictions.

```
AVG - RF MSE: 0.0001, XGB MSE: 0.0001
Training model for OBP...
OBP - RF MSE: 0.0002, XGB MSE: 0.0001
Training model for SLG...
SLG - RF MSE: 0.0007, XGB MSE: 0.0004
Training model for HR...
HR - RF MSE: 9.5712, XGB MSE: 4.0798
Training model for RBI...
RBI - RF MSE: 75.8823, XGB MSE: 38.3406
Training model for wOBA...
wOBA - RF MSE: 0.0002, XGB MSE: 0.0001
Training model for wRC+...
wRC+ - RF MSE: 92.0807, XGB MSE: 46.4729
```

The Mean Squared Error (MSE) results indicate that machine learning models outperform rate-based statistics like AVG, OBP, SLG, and wOBA with extremely narrow error margins, signifying high prediction accuracy for these metrics. In contrast, the models struggle more with counting numbers like HR and RBI, as well as scaled measures like wRC+, which have much higher MSEs—particularly the Random Forest model. XGBoost consistently outperforms Random Forest in all areas, lowering errors and displaying a stronger ability to manage data volatility and complexity. Overall, the findings demonstrate that while consistent offensive data may be projected accurately, more variable or context-sensitive statistics are more difficult to predict.

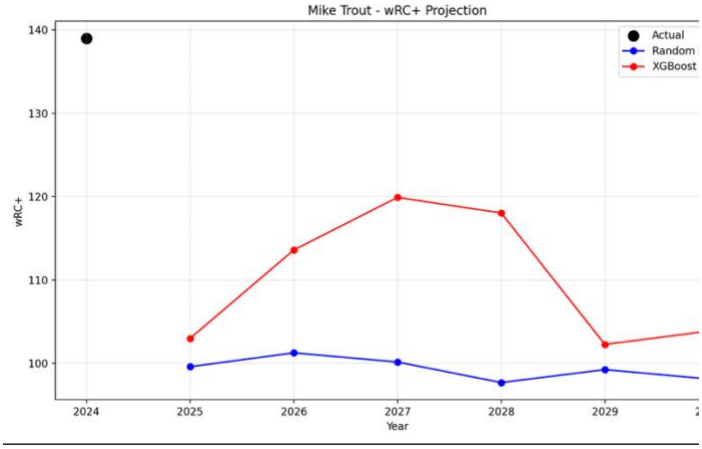
B. System Performance

To test the accuracy and performance of the system, I calculated predictions using the models on two different types of players: an established veteran, and guaranteed future Hall of Famer, and a young star who has already made a name for himself and is expected to have a historic career. For both players, I started in 2024 and predicted the upcoming 6 seasons.

Mike Trout (Established Veteran):

Projections for Mike Trout:

Year	Age	ProjectionType	AVG	OBP	SLG	HR	RBI	wOBA	
2024	32	Actual	0.220000	0.325000	0.541000	10	14	0.365000	139.
2025	33	RandomForest	0.229660	0.304630	0.444960	26.77	57.08	0.320420	99.
2025	33	XGBoost	0.244645	0.307804	0.514628	29.947689	71.221321	0.329807	103.
2026	34	RandomForest	0.229870	0.306980	0.442740	26.39	57.59	0.320080	101.
2026	34	XGBoost	0.237437	0.305514	0.514532	27.766409	68.126541	0.340006	113.
2027	35	RandomForest	0.226890	0.309830	0.444840	26.39	56.89	0.324800	100.
2027	35	XGBoost	0.239847	0.305898	0.514532	29.06654	69.543884	0.342522	119.
2028	36	RandomForest	0.226000	0.307840	0.444000	26.53	57.52	0.320730	97.
2028	36	XGBoost	0.232125	0.298888	0.514211	29.783241	67.554619	0.342962	118.
2029	37	RandomForest	0.225910	0.306310	0.440980	26.23	56.81	0.321630	99.
2029	37	XGBoost	0.229529	0.287915	0.526546	30.053133	71.916801	0.319125	102.
2030	38	RandomForest	0.225110	0.306310	0.440900	26.23	55.82	0.319880	98.
2030	38	XGBoost	0.229529	0.290098	0.526546	30.25399	71.916801	0.312511	103.

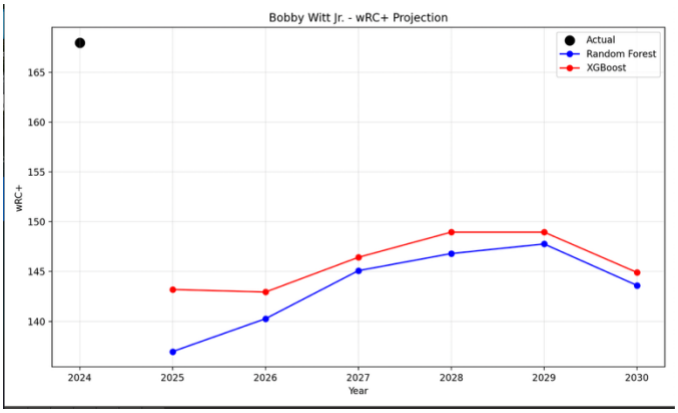


From the model predictions, we can see that the data follows expected trends. For an established veteran who has been plagued with injuries for most of his career, we see that Mike Trout is predicted to begin the decline in performance as he ages into his 38 year old season. This is expected of someone of this age and especially one where injuries are always a factor in every season.

Bobby Witt Jr (Breakout Star)

Projections for Bobby Witt Jr.:

Year	Age	ProjectionType	AVG	OBP	SLG	HR	RBI	wOBA	wRC+
2024	24	Actual	0.332000	0.389000	0.588000	32	109	0.410000	168.000000
2025	25	RandomForest	0.299660	0.390120	0.542130	29.83	86.1	0.386610	136.960000
2025	25	XGBoost	0.292341	0.376805	0.555291	29.525301	81.191757	0.372853	143.195847
2026	26	RandomForest	0.301710	0.390600	0.537830	29.56	85.14	0.386230	140.260000
2026	26	XGBoost	0.299983	0.378787	0.544358	28.990555	86.911087	0.377916	142.941071
2027	27	RandomForest	0.301840	0.391720	0.537870	29.43	86.21	0.386100	145.090000
2027	27	XGBoost	0.296198	0.378805	0.525941	27.082935	86.911087	0.381037	146.430267
2028	28	RandomForest	0.301660	0.392090	0.533280	28.73	85.31	0.387860	146.810000
2028	28	XGBoost	0.301358	0.380547	0.535157	27.082935	86.118233	0.383231	148.960449
2029	29	RandomForest	0.301020	0.391330	0.531860	28.64	84.64	0.388810	147.770000
2029	29	XGBoost	0.301358	0.381716	0.541514	27.094402	86.188576	0.383231	148.960449
2030	30	RandomForest	0.300830	0.390450	0.531800	28.79	83.45	0.384860	143.610000
2030	30	XGBoost	0.302288	0.376587	0.526767	23.597046	88.105377	0.381464	144.911758



For the breakout star in Bobby Witt Jr, we see that the model predictions also follow expected trends for a player of this nature. Bobby Witt Jr. is one of the greatest players currently in the league, has showcased his 5-tool abilities as a player and is expected to be one of the faces of the MLB for years to come. This predicted data shows that Witt Jr. continues his hot streak as expected, hitting his prime at a young age and continuing to put up all-star caliber numbers.

V. Conclusion and Future Work

This project clearly demonstrates the possibility of forecasting MLB player statistics with machine learning. Using historical data and robust models, I was able to obtain reasonably accurate projections for several significant batting stats.

Key Findings:

- Random Forest and XGBoost both made reliable predictions with small error margins.
- Using predictions as features in future projections allows for accurate multi-year forecasts.
- Front offices and fantasy players can use the tool to make data-driven decisions.

Future Directions:

- Additional features such as Statcast measurements (exit velocity, launch angle),

minor league performance, and physical traits may help increase prediction accuracy.

- Creating complementary models to estimate injury risk and playing time would result in more thorough projections.
- Incorporate pitching data to make predictions on both standard statistics like batting average, home runs, and rbis, as well as advanced statistics
- Build an app or web-based interfaced for widespread use and accessibility

VI. CONTRIBUTIONS

This project was completed independently, all components of this project were done by myself.